Sidharth Sreeram

Canadian Citizen | TN Visa Eligible (USMCA Engineer)
(416) 564-8590 | sidharthsreeram@gmail.com | sidharth-s.com | linkedin.com/in/sidharthsreeram | github.com/Sidharth-82

University of Waterloo Mechatronics Engineering graduate specializing in robotics, perception, and localization. Built real-time vision pipelines, EKF sensor fusion, and TDoA positioning across seven co-op and research terms.

Technical Skills

Programming Languages: Python, C, C++, MATLAB, SQL, Bash, Ladder Logic

Frameworks & Libraries: PyTorch, NumPy, OpenCV, Boto3, Pandas, Flask, ROS2, Nav2, Gazebo

Software: REST API, Backend Services, Cloud Computing, Database Management, Automated Testing

AI & ML: Reinforcement Learning, Multimodal Data Curation, Synthetic Data, Data Pipelines, VLA, VLM

Robotics: Object Detection, Segmentation, SLAM, RGB-D, LiDAR, Point Cloud, Sensor Fusion, Path Planning

Development: Linux, AWS, Claude Code, Git, LaTeX, Markdown, JSON, VSCode, Docker, Jira, Agile

Education

University of Waterloo | *B.ASc in Mechatronics & Artificial Intelligence* | *GPA: 3.8* 2021 – 2026

- **Relevant Courses:** Multivariable Control Systems, Control Applications, Cooperative and Adaptive Algorithms, Machine Learning, Autonomous Mobile Robotics, Autonomous Vehicles

Experience

VLA Research Assistant - UW WISE Lab (Research) | *Python, PyTorch* Jan 2026 – Apr 2026

- Researched and developed **Python** logic to organize visual and text data into structured VLM prompt format.
- Aggregated NuScenes-QA, NuCaptions, and NuX datasets with multi-sensor metadata by sample token.
- Reconstructed the **OpenDriveVLA** Stage-2 prompt spec, fusing camera, LiDAR, and CAN-bus ego state.
- Generated **728K** JSONL instruction-tuning samples for camera-vs-LiDAR VLA reliability research.

Software Analyst - Tamaki Controls (Coop) | *Python, Java, SQL, Ignition* May 2025 – Aug 2025

- Designed an **API** response schema improving cross-service communication between **Java** and **Python** stack.
- Built 6 customer-facing **REST** endpoints importing and exporting UTF-8 encoded CSV/JSON data.
- Wrote **SQL** queries aggregating source data from the database backing each export endpoint.
- Updated **API documentation** with input/output descriptions and example Python client scripts.

Platform Software Engineer - Ford (Coop) | *Python, C++, GCov* Sept 2024 – Dec 2024

- Developed a **Python CLI** for the AUTOSAR State Management API in a containerized Linux environment.
- Automated 6-stage state transitions, replacing stage-by-stage manual navigation with one command.
- Drove CLI adoption across **3 engineering teams** for state control in testing and integration.
- Created C++ test suites with GCov, raising line coverage **45% to 94%** and condition coverage **50% to 85%**.

Controls System Engineer - Brock Solutions (Coop) | *Python, SQL, Ignition* Jan 2024 – Apr 2024

- Built **Ignition SCADA HMI** screens visualizing live coil crane, acid bath, and pump operations at a steel facility.
- Simulated PLC outputs across a **27-PLC** system to validate HMI designs pre-commissioning.
- Collaborated in an **Agile** workflow using **Azure DevOps** to manage and track user stories and story points.

Product Developer - Gen AI Ventures (Coop) | *Flutter, Database Management* May 2023 – Aug 2023

- Collaborated in startup environment with founder and CPO to architect a social matching application using Flutter.
- Built responsive **Flutter** front-end with a reusable widget/component structure for adaptive UI across screens.
- Structured the Firebase backend: OAuth for sessions, Firestore for database, Cloud Functions for serverless logic.

Process Engineering Student - Matcor (Coop) | *Data Management, Excel* Sept 2022 – Dec 2022

- Gathered and structured use-case data of tooling across plant operations to evaluate equipment against project specs.
- Built Excel analyses turning the data into recommendations: commission new tooling or reuse existing machines.
- Conducted observational studies to obtain workflow data, resulting in enhancements to process efficiency and outputs.

Embedded Software Developer - Peraso (Coop) | *C/C++, Automated Testing* Jan 2022 – Apr 2022

- Developed and refactored embedded C++ firmware on an MCU managing serial flash for pre-production 5G silicon.
- Owned the flash-specific parser converting hex constant commands into executable bit sequences via bit-shifting.
- Led root cause analysis finding duplicate parsing functions; cutting write latency 5ms→1.3ms and read 2ms→1ms.

Projects

Cloud-Native ADAS Perception Stack | *Python, VLM, AWS, CARLA, Docker, boto3, KITTI, Synthetic Data*

- **Purpose**: Deployment study testing whether cloud-served detections stay viable in near real-time scenarios.
- **Completed:** Generated a 8,400-frame, 14,396-object **KITTI** dataset in **CARLA** (RGB, depth, instance-seg, LiDAR, ego pose): 14 highway scenes, 3 maps, day/night, mixed traffic density; test split held out on an unseen map.
- Held **GPU** spend to single-digit dollars per data collection cycle by rendering headless on **Amazon EC2** g4dn.xlarge spot instance and running CPU-intensive tasks on-premise.
- Built a bounded-memory **Amazon S3** streaming loader (thread-pool prefetch, decode-in-stream) so the full dataset processes on a laptop with flat memory and zero disk staging.
- Baked a custom AMI and dockerized programs to cut environment relaunch to ∼2 min and ensuring reproducibility.
- **Planned**: Detector training and benchmarking (mAP, MOTA/IDF1), cloud training with MLflow, ONNX endpoint benchmarked vs. Raspberry Pi edge, and closed loop demo.
- **AI Assisted Workflow**: Developed spec-first with Claude Code as an AI pair programmer, iterating each phase's design doc and implementation through rounds of critique and revision before committing code.

Self-Balancing Pendulum RL Control System | *C++, Reinforcemnt Learning, LibTorch, MuJoCo, Python, CMake*

- Built a C++ simulator, controller, and trainer for a motorized cart on a 2m rail balancing a free-swinging pole, where the only available control is accelerating the cart.
- Derived and implemented the pendulum physics from the equations of motion, running at 500Hz and modelling real motor force limits, sensor resolution and delay, and lost motor steps.
- Wrote the reinforcement learning algorithm (PPO) in C++ on LibTorch, training the neural network controller entirely on a laptop CPU.
- Trained a single policy that swings the pole upright from hanging and holds it there, reaching 96% of a perfect score, upright in 0.40 seconds and held within 0.03 degrees.
- Visualized network controller on MuJoCo, a physics engine, to verify reliability of the controller in real-world setting.

Machine Vision Path Following Robot | *Python, OpenCV, YOLO, Pure Pursuit, Raspberry Pi 5*

- Built a real-time **OpenCV** vision pipeline on embedded hardware for colour-based detection and tracking.
- Calibrated **RGB** camera intrinsics with chessboard targets and corrected lens distortion for accurate frames.
- Applied a homography perspective transform to map camera view to real-world top-down coordinates.
- Tuned HSV thresholding, morphological filtering, and contour extraction to isolate the path under varying lighting.
- Computed a look-ahead point via **Pure Pursuit**, deriving steering curvature regulated by a **PID** loop.
- Proposed and tested YOLOv8 object detection model to notify upon locating the target.

UWB Tag & Anchor Tracking System (Capstone) | *C++, ESP32-S3, PCBA, TDoA, RF Design, Localization*

- Developed **TDoA** localization using **weighted least squares** with **Firefly** optimization, reaching roughly **30 cm** 3-D accuracy in deployment.
- Implemented wireless clock synchronization for nanosecond-level timestamp reliability across the network.
- Designed and fabricated anchor PCBs integrating the **DW3110 UWB** transceiver and **ESP32-S3** MCU.
- Routed controlled-impedance RF traces, then brought up and debugged boards for reliable high-frequency signaling.

Mars Rover GPS Navigation System | *C++, Python, ROS2, Nav2, GPS, EKF, Docker, Gazebo, SLAM*

- Engineered a **GPS** waypoint-following navigation stack with **ROS2** and **Nav2** for autonomous traversal of terrain.
- Fused GPS, IMU, and wheel odometry using dual **EKFs** in robot_localization for drift-corrected global localization.
- Integrated **Azure Maps tiles** into Mapviz for terrain and elevation overlays that informed global path planning.

Certifications

AWS Certified Cloud Practitioner | *AWS Educate - In Progress* — June 2026 – Present

AWS Cloud 101 | *AWS Educate* — August 2026

Databricks Fundamentals | *Databricks Academy* — July 2026

ROS 2 Nav2 - SLAM and Navigation | *Udemy* — July 2026